

Time Series Exercise Sheet 6

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Exercise 6.1

Let $g \in L^1$, have Fourier transform denoted by $\mathcal{G}$. Prove the following relations (where $\mathcal{H}$ is taken to be the Fourier transform of h).

1. If $h(t) = g(t)^*$ for all $t \in \mathbb{R}$ then

$$\mathcal{H}(f) = \mathcal{G}(-f)^*$$

for all $f \in \mathbb{R}$.

2. If $h(t) = g(\alpha t)$ for all $t \in \mathbb{R}$ then

$$\mathcal{H}(f) = \mathcal{G}(f/\alpha)/|\alpha|$$

for all $f \in \mathbb{R}$.

3. If $h(t) = g(t + \tau)$ for all $t \in \mathbb{R}$ then

$$\mathcal{H}(f) = \mathcal{G}(f)e^{2\pi i f \tau}$$

for all $f \in \mathbb{R}$.

4. Let $p \in L^1$ with Fourier transform $\mathcal{P}$. If $h(t) = \alpha g(t) + p(t)$ for all $t \in \mathbb{R}$ then

$$\mathcal{H}(f) = \alpha \mathcal{G}(f) + \mathcal{P}(f)$$

for all $f \in \mathbb{R}$.

Solution 6.1

1. We have $h(t) = g(t)^*$

$$\begin{aligned} H(f) &= \int_{-\infty}^{\infty} h(t)e^{-2\pi i t f} dt \\ &= \int_{-\infty}^{\infty} g(t)^* e^{-2\pi i t f} dt \\ &= \int_{-\infty}^{\infty} g(t)e^{2\pi i t f} dt^* \\ &= G(-f)^*. \end{aligned}$$

2. We have $h(t) = g(\alpha t)$, then

$$\begin{aligned} H(f) &= \int_{-\infty}^{\infty} h(t)e^{-2\pi i t f} dt \\ &= \int_{-\infty}^{\infty} g(\alpha t)e^{-2\pi i t f} dt \\ &= \int_{-\infty}^{\infty} g(t')e^{-i \frac{f}{\alpha} t'} \frac{1}{|\alpha|} dt' \\ &= G(f/\alpha)/|\alpha| \end{aligned}$$

3. We have $h(t) = g(t + \tau)$, we have

$$\begin{aligned} H(f) &= \int_{-\infty}^{\infty} h(t) e^{-2\pi i f t} dt \\ &= \int_{-\infty}^{\infty} g(t + \tau) e^{-2\pi i f t} dt \\ &= \int_{-\infty}^{\infty} g(t + \tau) e^{-2\pi i f (t + \tau)} dt e^{2\pi i f \tau} \\ &= G(f) e^{2\pi i f \tau} \end{aligned}$$

4. This follows from linearity of integration.

Exercise 6.2

Prove the continuous time version of the convolution theorem, i.e. let $h, g \in L^1$, and let $u = g * h$, then

$$\mathcal{U} = \mathcal{G} \cdot \mathcal{H}.$$

Solution 6.2

We have for any $f \in \mathbb{R}$

$$\begin{aligned} U(f) &= \int_{-\infty}^{\infty} u(x) e^{-2\pi i x f} dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x - y) h(y) dy e^{-2\pi i x f} dx \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x - y) e^{-2\pi i x f} dx h(y) dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(s) e^{-2\pi i s f} ds h(y) e^{-2\pi i y f} dy \\ &= G(f) H(f). \end{aligned}$$

The interchange of integrals is justified because

$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g(x - y) h(y)| dx dy &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |g(s) h(y)| ds dy \\ &= \int_{-\infty}^{\infty} |g(s)| ds \int_{-\infty}^{\infty} |h(y)| dy \\ &< \infty \end{aligned}$$

by assumption, and so we may apply Fubini's theorem.

Exercise 6.3

Prove the following theorem. Consider a continuous function $g \in L^1$. Then writing $\mathcal{G}$ for its Fourier transform and G for the Fourier transform of the sequence $\{g(t)\}_{t \in \Delta\mathbb{Z}}$ and assuming that $\mathcal{G} \in L^1$ and $\{g(t)\}_{t \in \Delta\mathbb{Z}} \in \ell^1$ and the sequence $\{\mathcal{G}(f + k/\Delta)\}_{k \in \mathbb{Z}} \in \ell^1$ for all $f \in \mathbb{R}$. Then we have

$$G(f) = \sum_{k \in \mathbb{Z}} \mathcal{G}(f + k/\Delta). \quad (1)$$

Solution 6.3

Firstly, recall that we have the Fourier inverse results so that for any $t \in \Delta\mathbb{Z}$

$$\begin{aligned} g(t) &= \int_{-\infty}^{\infty} \mathcal{G}(f) e^{2\pi i f t} df, \\ g(t) &= \int_{-1/2\Delta}^{1/2\Delta} G(f) e^{2\pi i f t} df. \end{aligned}$$

Now notice that

$$\begin{aligned}
g(t) &= \int_{-\infty}^{\infty} \mathcal{G}(f) e^{2\pi i f t} \mathrm{d}f \\
&= \sum_{k \in \mathbb{Z}} \int_{k/\Delta - 1/2\Delta}^{k/\Delta + 1/2\Delta} \mathcal{G}(f) e^{2\pi i f t} \mathrm{d}f \\
&= \sum_{k \in \mathbb{Z}} \int_{-1/2\Delta}^{1/2\Delta} \mathcal{G}(f + k/\Delta) e^{2\pi i f t} \mathrm{d}f e^{2\pi i t k/\Delta} \\
&= \sum_{k \in \mathbb{Z}} \int_{-1/2\Delta}^{1/2\Delta} \mathcal{G}(f + k/\Delta) e^{2\pi i f t} \mathrm{d}f
\end{aligned}$$

because $t = z\Delta$ for some $\Delta \in \mathbb{Z}$ and $e^{2\pi i x} = 1$ for any $x \in \mathbb{Z}$. Now we may swap the order of summation and integration since

$$\sum_{k \in \mathbb{Z}} \int_{-1/2\Delta}^{1/2\Delta} |\mathcal{G}(f + k/\Delta)| \mathrm{d}f < \infty$$

by assumption. Therefore we have

$$\int_{-1/2\Delta}^{1/2\Delta} \sum_{k \in \mathbb{Z}} \mathcal{G}(f + k/\Delta) e^{2\pi i f t} \mathrm{d}f = g(t) = \int_{-1/2\Delta}^{1/2\Delta} G(f) e^{2\pi i f t} \mathrm{d}f.$$

It therefore follows that

$$\sum_{k \in \mathbb{Z}} \mathcal{G}(f + k/\Delta) = G(f)$$

for almost every $f \in [-1/2\Delta, 1/2\Delta]$. Finally, we have that G is continuous because $\{g(t)\}_{t \in \Delta\mathbb{Z}} \in \ell^1$. The sum on the right hand side is also continuous because

$$\sum_{k \in \mathbb{Z}} |\mathcal{G}(f + k/\Delta)| < \infty$$

by assumption. Therefore, the result holds for all $f \in [-1/2\Delta, 1/2\Delta]$.